231501053

Gokulakkannan P

Generative AI

ex.no:06

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```
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Downloading faiss_cpu-1.13.2-cp310-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (23.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 23.8/23.8 MB 30.0 MB/s eta 0:00:00

Installing collected packages: faiss-cpu
Successfully installed faiss-cpu-1.13.2
Total vectors indexed: 8

Query: space and future survival

Rank 1: Interstellar (Distance: 1.2931)
Rank 2: The Matrix (Distance: 1.5961)
Rank 3: Inception (Distance: 1.6286)

Query: romantic love story

Rank 1: The Notebook (Distance: 1.0449)
Rank 2: Titanic (Distance: 1.1327)
Rank 3: John Wick (Distance: 2.0000)

Query: superhero battle

Rank 1: John Wick (Distance: 1.0000)
Rank 2: Inception (Distance: 1.0000)
Rank 3: The Dark Knight (Distance: 1.0000)
```

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```
# 4. Similarity Search Function
# ------------------------------
def search_similar(query, top_k=3):
    query_vec = vectorizer.transform([query]).toarray().astype("float32")

    distances, indices = index.search(query_vec, top_k)

    print(f"\nQuery: {query}\n")
    for i, idx in enumerate(indices[0]):
        print(f"Rank {i+1}: {df.iloc[idx]['title']} (Distance: {distances[0][i]:.4f})")

# ------------------------------
# 5. Test Search
# ------------------------------
search_similar("space and future survival")
search_similar("romantic love story")
search_similar("superhero battle")
```

```
Collecting faiss-cpu
  Downloading faiss_cpu-1.13.2-cp310-abi3-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (7.6 kB)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: numpy<3.0,>=1.25.0 in /usr/local/lib/python3.12/dist-packages (from faiss-cpu) (2.0.2)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from faiss-cpu) (26.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

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```
            Space exploration to save humanity ,
            "Dreams within dreams and mind-bending reality",
            "A vigilante fights crime in Gotham city"
        ]
    }

    df = pd.DataFrame(data)

    # -------------------------------
    # 2. Convert Text → Embeddings
    # -------------------------------
    vectorizer = TfidfVectorizer()
    X = vectorizer.fit_transform(df["description"]).toarray()

    # Convert to float32 (required for FAISS)
    X = np.array(X).astype("float32")

    # -------------------------------
    # 3. Build FAISS Index
    # -------------------------------
    dimension = X.shape[1]
    index = faiss.IndexFlatL2(dimension)   # L2 distance
    index.add(X)

    print(f"Total vectors indexed: {index.ntotal}")

    # -------------------------------
    # 4. Similarity Search Function
    # -------------------------------
    def search_similar(query, top_k=3):
```

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```
# Install required libraries
!pip install faiss-cpu pandas scikit-learn

import numpy as np
import pandas as pd
import faiss
from sklearn.feature_extraction.text import TfidfVectorizer

# -------------------------------
# 1. Sample Dataset (Movies)
# -------------------------------
data = {
    "title": [
        "The Matrix",
        "John Wick",
        "Avengers: Endgame",
        "Titanic",
        "The Notebook",
        "Interstellar",
        "Inception",
        "The Dark Knight"
    ],
    "description": [
        "A hacker discovers reality is a simulation and fights machines",
        "An assassin seeks revenge in a criminal underworld",
        "Superheroes unite to defeat a powerful villain",
        "A tragic love story on a sinking ship",
        "A romantic drama about enduring love",
```